

- (1) All questions are compulsory.
- (2) Make suitable assumptions wherever necessary and state the assumptions made.
- (3) Answers to the same question must be written together.
- (4) Numbers to the right indicate marks.
- (5) Draw neat labeled diagrams wherever necessary.
- (6) Use of non-programmable calculator is allowed.

Q 1 Attempt any TWO of the following:

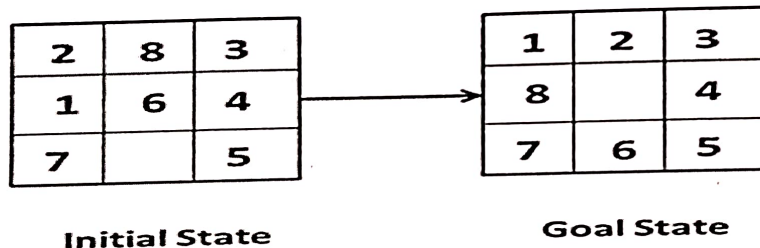
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- A. List and briefly explain the six types of foundational languages used in Artificial Intelligence.
- B. Explain in your own words the four different approaches to Artificial Intelligence (AI) based on their focus and perspective. Also, describe the Turing Test approach in detail and its significance in AI.
- C. Use real-world examples to explain different types of AI environments in which an agent operates.
- D. Analyze a learning-based AI agent and explain its four conceptual components. Draw a diagram to illustrate how these components interact and justify the role of each component in the agent's learning process.

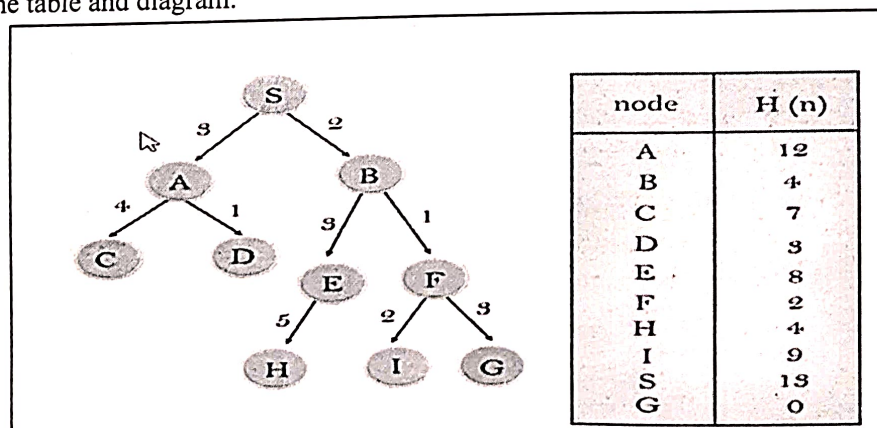
Q 2 Attempt any TWO of the following:

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- A. Examine the n-queen problem and break down the components required to formulate it. Describe how each component contributes to finding a solution.
- B. Evaluate the following scenario and formulate it as an 8-puzzle problem. Identify the initial state, goal state, possible actions and path cost and explain the steps required to reach the goal state.



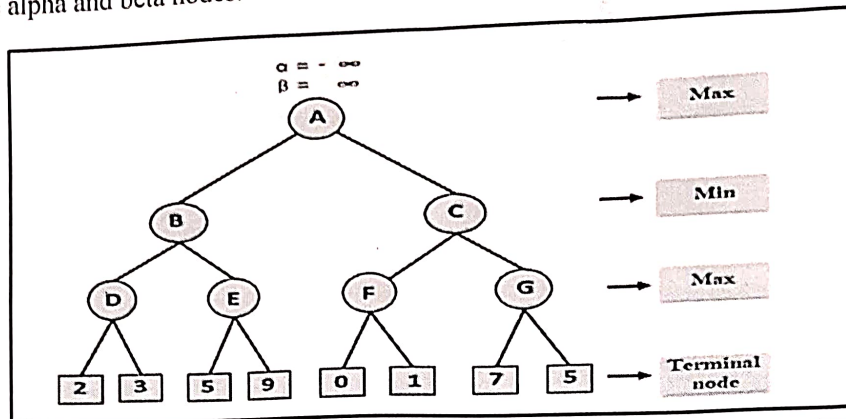
- C. Given the following search problem, use the Greedy Best-First Search algorithm to traverse the graph. Show the open list, closed list and the final solution path from the Start State S to the Goal Node G, using the heuristic values provided in the table and diagram.



- D. List the different types of online search agents used in unknown environments and recall how each type functions with examples.

Q 3 Attempt any TWO of the following:

- A. Explain the different techniques used to find the best optimal solution in problem-solving. Also, describe the key elements of game-playing search and their roles.
- B. Define Alpha-Beta Pruning. Using the game tree provided below, apply the alpha-beta pruning algorithm to evaluate it and determine the cutoff values for the alpha and beta nodes.



- C. Examine the two types of inference systems in Knowledge-Based Agents (KBA). Explain how each system works and compare their differences in terms of reasoning approach and functionality.
- D. List and explain the PEAS (Performance measure, Environment, Actuators, Sensors) descriptors for the WUMPUS World.

Q 4 Attempt any TWO of the following:

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- A. Define quantifiers in logic. List and explain their different types with syntax and examples.
- B. Explain the concepts of unification and lifting. Describe the process of unification and lifting. Apply unification to the following expressions and show the substitutions (i) $P(X, a)$ and $P(b, Y)$; (ii) $\text{likes}(\text{john}, X)$ and $\text{likes}(Y, \text{mary})$.
- C. Compare and contrast Forward and Backward chaining reasoning methods.
- D. Convert the following facts into First-Order Logic (FOL) and use Backward-Chaining to prove that "Colonel is a Criminal":
 - (i) It is a crime for an American to sell weapons to the enemy of America.
 - (ii) Country Nono is an enemy of America.
 - (iii) Nono has some missiles.
 - (iv) All the missiles were sold to Nono by Colonel.
 - (v) Missile is a weapon.
 - (vi) Colonel is American.

Q 5 Attempt any TWO of the following:

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- A. Write a short note on the following planning methods (i) Total Ordered Planning; (ii) Partial Ordered Planning.
- B. Provide a brief overview of Planning Graphs. Apply this knowledge to analyze planning approaches, considering factors such as time, scheduling and resources. Also explain why these considerations are necessary in classical planning.
- C. Explain the concept of conditional planning and illustrate with an example how it helps in decision-making under uncertainty.
- D. Evaluate the Internet Shopping World and discuss its importance, advantages and potential drawbacks for users and businesses.